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Structural Information as the Missing Component in the Radio–Star Formation Relation

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ABSTRACT

The infrared-radio correlation carries $\sigma \sim 0.2$–0.3 dex of scatter whose physical origin has remained unidentified. We quantify galaxy structure as a contributing hidden variable using two independent morphological systems applied to VLA-COSMOS 3 GHz sources. A five-seed information ladder ($N = 4843$) shows non-parametric morphology (Gini, concentration, asymmetry) delivers $\Delta R^2 = 0.09 \pm 0.03$, exceeding the noise floor of every other variable by a factor of five and $2.8 \pm 1.2\times$ the combined contribution of all other variables. The gain is indistinguishable from zero at low stellar mass, grows large at high stellar mass ($\Delta R^2 = 0.19 \pm 0.04$), and vanishes when radio-excess hosts are removed, pointing to two channels: AGN/radio-excess identification and a smaller calorimetric channel, corroborated by a surface-density ratio correlating with morphology 3–6$\times$ more strongly than SFR surface density alone. Galaxy Zoo Hubble citizen-science classifications independently replicate the signal (AUC 0.697 ± 0.009 vs. 0.693 ± 0.009), with a retention fraction after redshift control (0.51 ± 0.03) statistically indistinguishable from Zurich's (0.49 ± 0.24). Matched-sample tests show radio-excess hosts are more compact (Gini $d = +0.28$; concentration $d = +0.27$) and smoother (asymmetry $d = -0.24$; Galaxy Zoo Hubble spiral $d = -0.369$) than controls, with all nine Zurich tests surviving Holm–Bonferroni correction. Flat-spectrum sources ($N_{\rm rx} = 139$) show extreme asymmetry suppression ($d = -0.87$), surviving spectral-index threshold cuts. A WISE colour-colour cross-check on multiwavelength AGN classification shows 75.8% agreement. Adding morphology to a nested logistic model does not measurably improve full-sample radio-excess discrimination beyond mass, redshift, and SFR (ΔAUC $= -0.0002$; 95% CI includes zero). Non-parametric morphology is the strongest structural discriminator for radio-excess identification; a single smooth/featured classification from Euclid, Rubin LSST, or Roman will improve radio-excess screening in the SKA era.

Keywords: galaxies: star formation — galaxies: structure — radio continuum: galaxies — galaxies: evolution — galaxies: active

1. INTRODUCTION

Radio continuum emission is one of astronomy's most reliable star-formation tracers because it penetrates dust. Synchrotron radiation from cosmic-ray (CR) electrons accelerated in supernova remnants, combined with thermal free-free emission from ionised gas, traces recent massive star formation where optical photometry fails (Condon 1992). The empirical infrared-radio correlation (IRRC), linking radio luminosity to total infrared luminosity and thence to SFR, has underpinned decades of cosmic star-formation census work (Helou et al. 1985; Yun et al. 2001; Bell 2003; Murphy et al. 2011; Madau & Dickinson 2014). The IRRC is not a perfect one-parameter mapping. Its $\sigma \sim 0.2$–0.3 dex scatter encodes genuine physics: the synchrotron yield per unit SFR depends on (i) the calorimetric fraction, which rises with gas surface density and infrared luminosity (Lacki et al. 2010); (ii) the magnetic field strength, setting the synchrotron cooling timescale (Schleicher & Beck 2013); (iii) inverse-Compton losses off the cosmic microwave background at high redshift (Lacki et al. 2010; Schober et al. 2016); and (iv) the free-free fraction, varying with frequency and ionisation

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state (Condon 1992). Every one of these processes is regulated by structure: a compact, high-surface-density system radiates more synchrotron per unit SFR than a diffuse irregular of the same mass – but, as we show below, only once that system is close enough to the calorimetric limit for the effect to dominate over other sources of scatter.

Observational evidence for this structural dependence has grown. Magnelli et al. (2015) found a trend of $q_{\rm IR}$ with stellar mass. Delhaize et al. (2017) detected redshift evolution of $q_{\rm IR}$, and Delvecchio et al. (2021) showed that $q_{\rm IR}$ depends jointly on mass and redshift consistent with inverse-Compton losses dominating at high z. These results establish that the IRRC is not universal, but the structural degrees of freedom driving the scatter require a controlled, cross-validated framework that separates a calorimetric effect within the star-forming population from a signature of AGN/radio-excess contamination. Distinguishing these two possibilities – which we refer to throughout as two separate *channels* – is a central goal of this work.

A second practical complication is AGN contamination: radio-excess sources (Del Moro et al. 2013) must be identified structurally to avoid biasing radio-SFR calibrations for next-generation surveys. Non-parametric morphology offers a principled route to both problems. The Gini coefficient (Abraham et al. 2003), the concentration-asymmetry system (Conselice 2003), and the Gini-M_{20} framework (Lotz et al. 2004) encode compactness, nuclear dominance, disturbance, and clumpiness without parametric assumptions. Concentration is a direct proxy for the gas surface densities regulating CR calorimetric efficiency (Thompson et al. 2006; Lacki et al. 2010); asymmetry traces dynamical disturbance long proposed as an AGN trigger (Lotz et al. 2008; Ellison et al. 2013); Gini captures nuclear light concentration. Crucially, these statistics are computed directly from the pixel flux distribution without assuming any functional form for the light profile, making them sensitive to substructure (nuclear point sources, asymmetric tidal features, star-forming clumps) that a smooth parametric model cannot capture.

Citizen-science classifications from Galaxy Zoo (Lintott et al. 2008; Willett et al. 2013) provide an independent, scalable route to the same structural information. If the morphological signal is real, it should appear in volunteer vote fractions derived from a completely different measurement process, and it should behave consistently under the same robustness tests applied to the pixel-level data. Makechemu et al. (2026) establish the complementary bound: six regression architectures spanning linear to deep non-linear models all saturate at $R^2 \lesssim 0.45$ from radio luminosity alone, confirming the dominant scatter is physical rather than methodological.

The COSMOS field (Scoville et al. 2007) provides the testbed for this work. The VLA-COSMOS 3 GHz Large Project (Smolčić et al. 2017) delivers $\sim$2.3 μJy beam^{-1} sensitivity at 0.″75 resolution over $\sim$2 deg^2, with multiwavelength AGN classification (Delvecchio et al. 2017; Novak et al. 2017). The Zurich/Tasca HST/ACS non-parametric catalogue (Tasca et al. 2009) and the Galaxy Zoo Hubble citizen-science catalogue (Willett et al. 2017) both cover this field at sufficient depth for a controlled comparison.

We pursue four questions: (1) what is the incremental predictive hierarchy of physical information beyond radio luminosity, and how robust is that hierarchy to the stochasticity of the machine-learning model used to measure it? (2) which morphological parameters carry the strongest independent signal for radio-excess behaviour? (3) does a citizen-science classification system recover the same structural signal as precision HST measurements, with the same seed-to-seed stability? (4) is the morphology gain a single physical effect, or does it decompose into an AGN-identification component and a separate calorimetric component isolable by removing radio-excess hosts and stratifying on stellar mass? Question 1 is answered in Section 4.1 and its multi-seed extension (Section 4.1.1), and question 4 in Sections 4.2–5.3.

2. DATA AND SAMPLE

2.1. *VLA-COSMOS 3 GHz Catalogue*

We draw our parent sample from the VLA-COSMOS 3 GHz Large Project catalogue (Smolčić et al. 2017), extracting sky positions, photometric redshifts from COSMOS2015 (Laigle et al. 2016), stellar masses and SF-component infrared luminosities ($\log L_{\rm TIR,SF}$) from SED fitting (Delvecchio et al. 2017), rest-frame radio luminosities at 1.4 and 3 GHz, multiwavelength AGN flags, and the radio-excess flag. After quality cuts, $N = 7729$ sources remain.

2.2. *Zurich/Tasca Morphology Catalogue*

The COSMOS morphology catalogue (Tasca et al. 2009), constructed from HST/ACS I-band imaging at 0.″03 pix^{-1} resolution, provides the Gini coefficient (Abraham et al. 2003), concentration and asymmetry (Conselice 2003), half-light radius $R_{\rm HALF}$, axial ratio q, and an SVM-based morphological class. We convert $R_{\rm HALF}$ to physical half-light

Table 1. Fixed Matched Sample Properties

Quantity	Value
Sample sizes	
VLA working sample (quality cuts)	7729
Zurich catalogue (coord. cleaning)	237,912
Zurich matched within $0\farcs7$	4844
Fixed ladder sample (N)	4843
GZ Hubble matched within $0\farcs7$	3696
Medians (fixed sample)	
Redshift z	0.91
$\log(M_\star/M_\odot)$ ($M_\odot$)	10.87
$\log L_{1.4\,\mathrm{GHz}}$ ($\mathrm{W\,Hz^{-1}}$)	23.20
$\log L_{3\,\mathrm{GHz}}$ ($\mathrm{W\,Hz^{-1}}$)	22.97
R_e (kpc)	2.51
Gini / Concentration / Asymmetry	0.540 / 0.342 / 0.086
Activity classes (binary radio-excess flag)	
Non-radio-excess (N_non)	3840
Radio-excess (N_rx)	1003

radius R_e under the Planck 2018 cosmology (Planck Collaboration et al. 2020), define a stellar surface-density proxy $\log\Sigma_\star \equiv \log(M_\star/M_\odot) - \log(2\pi R_\mathrm{e}^2)$, and an inclination proxy $i_\mathrm{proxy} \equiv \arccos(q)$.

2.3. *Galaxy Zoo Hubble*

The Galaxy Zoo Hubble catalogue (Willett et al. 2017) provides debiased and weighted volunteer vote fractions from non-expert visual classification of HST images. We retain 84,954 COSMOS sources and sky-match to the VLA catalogue within $0\farcs7$, yielding $N_\mathrm{match} = 3696$ (median separation 0.14 arcsec). The primary predictor is the debiased featured-or-disk fraction ($t01$), which corrects for the known bias that volunteers classify high-redshift galaxies as smooth at a rate exceeding their true structural smoothness (Bamford et al. 2009). Additional vote fractions used in the ladder are: odd features ($t06$), merger ($t08$), disturbed ($t08$), spiral ($t04$), bulge dominance ($t05$), edge-on probability ($t02$), and clumpy fraction ($t12$).

2.4. *Spectral Index, Compactness, and Sample Construction*

A two-band spectral index is computed as $\alpha_\mathrm{radio} = (\log L_{3\,\mathrm{GHz}} - \log L_{1.4\,\mathrm{GHz}})/\log_{10}(3.0/1.4)$. Because both luminosities were derived under a fixed $\alpha = -0.7$ (Delvecchio et al. 2017), α_radio has limited dynamic range (median $= -0.700$) and should be regarded as a qualitative indicator of emission character. We use it to define flat-spectrum ($\alpha_\mathrm{radio} \geq -0.5$; $N = 278$; 5.7%) and steep-spectrum ($\alpha_\mathrm{radio} < -0.5$; $N = 4565$; 94.3%) sub-samples. An independent geometry-based compactness proxy, the radio-to-optical size ratio $\mathrm{ROR} \equiv \log_{10}[\theta_\mathrm{beam}/(2\sqrt{2\ln 2}\cdot R_{\mathrm{HALF},''})]$ with $\theta_\mathrm{beam} = 0\farcs75$, agrees with the spectral-index classification for 52.3% of sources, confirming the two proxies measure different physical dimensions. Sky-matching VLA and Zurich catalogues within $0\farcs7$ yields $N = 4844$ sources; enforcing complete coverage in all ladder variables gives the *fixed matched sample* of $N = 4843$ galaxies used throughout (Table 1). All ladder experiments use this fixed sample, so performance differences between rungs cannot reflect changes in sample composition.

3. METHODS

3.1. *Information Ladder*

Our regression target is $\log L_{\mathrm{TIR,SF}}$. We define cumulative feature sets in which each rung retains all features from lower rungs. For the Zurich analysis the eight rungs are: (1) $L_{1.4\,\mathrm{GHz}}$ only; (2) $+L_{3\,\mathrm{GHz}}$; (3) $+\log(M_\star/M_\odot)$; (4) $+R_{\mathrm{e}}$; (5) $+\log\Sigma_\star$; (6) + Gini, concentration, asymmetry; (7) $+i_{\mathrm{proxy}}$; (8) + `CLASS_SVM`. The Galaxy Zoo Hubble ladder adds each of its eight vote fractions sequentially from the same radio-mass baseline. The primary inference model is a multilayer perceptron (MLP; two hidden layers of 64 and 32 neurons, ReLU activations, Adam optimisation, early stopping on a 10% validation split). All results use 5-fold cross-validation.

3.1.1. *Multi-Seed Robustness Protocol*

A single MLP training run carries irreducible stochasticity from both its weight initialisation and its early-stopping validation split, in addition to the cross-validation fold assignment. To obtain estimates that are representative of the model's typical behaviour, we repeat every headline ladder computation across five independent seeds (7, 13, 21, 42, 99), varying both the model's internal random state and the cross-validation split together, and report the resulting mean, standard deviation, and range. We adopt the following standard: a rung's ΔR^2 is treated as a genuine, reportable signal only if it is consistently signed across all five seeds and exceeds the empirical seed-to-seed noise floor established by the other rungs in the same ladder. This protocol is applied identically to the Zurich and Galaxy Zoo Hubble ladders, to the redshift-sensitivity check on both, and to the AGN-free/radio-excess-free and mass-stratified sub-analyses (Sections 4.2–4.3).

3.1.2. *Redshift-Sensitivity Check*

A redshift-sensitivity check is applied to both ladders: redshift is inserted explicitly after stellar mass to quantify how much of the morphological ΔR^2 reflects redshift-correlated trends rather than genuine structural information. The resulting *retention fraction* (the ratio of ΔR^2 with z controlled to ΔR^2 without) is computed under the same five-seed protocol as every other quantity in this paper.

3.2. *Radio-Excess Classification and Matched-Sample Tests*

We evaluate logistic-regression ROC-AUC (5-fold stratified) for individual and joint morphology parameters, and nested models extending $\{\log(M_\star/M_\odot), z, \log L_{\mathrm{TIR,SF}}\}$. BIC from maximum-likelihood fits provides penalised model selection. For the nested comparison we additionally report a bootstrap confidence interval on ΔAUC (full model minus base model), obtained by resampling the out-of-fold predicted probabilities from both models 2000 times, since the base and full model AUCs agree closely despite a nonzero BIC penalty (Section 4.8).

The central test is a **unique one-to-one nearest-neighbour matched-sample comparison**: each radio-excess source is matched to its closest non-radio-excess counterpart in standardised $(\log(M_\star/M_\odot), z, \log L_{\mathrm{TIR,SF}}, \log L_{1.4\,\mathrm{GHz}})$ space with no replacement, producing $N_{\mathrm{rx}} = N_{\mathrm{non}} = 1003$ independent pairs (Zurich) or 697 pairs (Galaxy Zoo Hubble). Matching additionally on $\log L_{1.4\,\mathrm{GHz}}$ ensures that the morphological offsets reported below cannot be driven by differences in radio luminosity between radio-excess hosts and their controls. This is a deliberately conservative choice: because $L_{\mathrm{TIR,SF}}$ and $L_{1.4\,\mathrm{GHz}}$ are themselves coupled through the infrared-radio correlation, controlling for both simultaneously removes any residual structural signal that merely tracks the part of $L_{1.4\,\mathrm{GHz}}$ correlated with star formation, leaving only the morphological offset that is independent of both star-formation rate and radio luminosity. Four-variable matching strengthens rather than weakens the offsets relative to a three-variable $(\log(M_\star/M_\odot), z, \log L_{\mathrm{TIR,SF}})$ match, confirming the signal is not an artefact of radio-luminosity differences. Morphological differences are quantified by Cohen's d with 95% bootstrap confidence intervals (5000 resamples for Zurich; 2000 for Galaxy Zoo Hubble). Cohen's d is the mean difference between radio-excess hosts and matched controls in units of the pooled standard deviation, $d = (\bar{x}_{\mathrm{rx}} - \bar{x}_{\mathrm{non}})/\sqrt{(\sigma^2_{\mathrm{rx}} + \sigma^2_{\mathrm{non}})/2}$; a positive value indicates the radio-excess population has the higher mean, and $|d| = 0.2$, 0.5, and 0.8 correspond to the conventional small, medium, and large effect-size thresholds. We pair the standardized effect size with a distribution-free Kolmogorov-Smirnov test and with bootstrap confidence intervals, so the inference makes no Gaussianity assumption.

Because multiple matched-sample tests are reported within each family (three morphology metrics across three spectral/compactness sub-samples for Zurich; six vote fractions for Galaxy Zoo Hubble), we apply Holm–Bonferroni correction for multiple comparisons within each family and report which tests survive it at family-wise $\alpha = 0.05$.

Table 2. Five-Seed Robustness Summary (seeds 7, 13, 21, 42, 99)

System	N	Primary rung ΔR^2	Retention fraction	Model
Zurich (Gini, Conc., Asym.)	4843	0.092 ± 0.028 $[0.060, 0.127]$	0.49 ± 0.24 $[0.28, 0.91]$	MLP
Galaxy Zoo Hubble (Feat./Disk)	3696	0.153 ± 0.007 $[0.141, 0.160]$	0.51 ± 0.03 $[0.47, 0.55]$	MLP

Note—All values are five-seed mean ± standard deviation, with the range in square brackets. Zurich's retention fraction is too imprecise to distinguish from Galaxy Zoo Hubble's, given the overlapping uncertainties (Section 4.1.2).

3.3. *Two-Channel Diagnostics*

To test whether the morphology gain reflects a single physical mechanism or decomposes into distinct channels, we apply three further diagnostics, each under the five-seed protocol above. First, we recompute the full ladder after removing (a) all multiwavelength-flagged AGN and (b) all radio-excess sources, testing whether the morphology gain survives the removal of the population it may exist to identify. Second, we stratify the fixed ladder sample into stellar-mass terciles and recompute the morphology-rung ΔR^2 within each bin, testing for mass dependence consistent with a calorimetric mechanism that engages more strongly as systems approach the calorimetric limit. Third, we construct a direct surface-density diagnostic, $\log(\Sigma_{L,L_{\mathrm{TIR,SF}}}/\Sigma_\star)$ where $\Sigma_{L,L_{\mathrm{TIR,SF}}} \equiv L_{\mathrm{TIR,SF}}/(2\pi R_{\mathrm{e}}^2)$, and correlate it with morphology via Spearman's ρ; and we compute permutation importance on the full-feature MLP, grouping $M_\star$, R_{e}, and $\log \Sigma_\star$ into a single permutation unit since $\log \Sigma_\star$ is a deterministic function of the other two.

As an independent cross-check on the multiwavelength AGN classification used throughout, we sky-match AllWISE photometry to the VLA-COSMOS sample ($2\overset{''}{.}0$ tolerance) and apply the Stern et al. (2012) $W1 - W2 > 0.8$ colour cut, comparing the resulting flag against the combined multiwavelength AGN flag.

4. RESULTS

4.1. *Two Systems, One Signal*

Table 3 presents the ladder for both systems at a representative seed (42); Table 2 presents the five-seed estimates used throughout the rest of this paper. Figure 1 shows the two ladders side by side.

Zurich non-parametric morphology. Starting from radio luminosity alone, the MLP achieves $R^2 = 0.465$ at the representative seed. Adding a second radio band, stellar mass, physical size, and surface density collectively recover modest additional variance. At rung 6, adding Gini, concentration, and asymmetry simultaneously produces $\Delta R^2_{\mathrm{morph}} = +0.127$ at this seed, raising cumulative R^2 from 0.495 to 0.621.

4.1.1. *Multi-Seed Robustness*

Because the MLP's own weight initialisation and early-stopping split add sampling variance on top of the cross-validation folds, we repeat the full eight-rung ladder across five seeds (7, 13, 21, 42, 99), following the protocol of Section 3.1.1. The morphology rung's ΔR^2 is positive under every seed (mean 0.092, standard deviation 0.028, range $[0.060, 0.127]$) and exceeds the pooled seed-to-seed standard deviation of every other rung's ΔR^2 (0.018) by a factor of 5.0. No other rung is robustly distinguishable from zero at this level: stellar mass (mean $\Delta R^2 = +0.002$, std 0.009), surface density (mean -0.004, std 0.006), and morphological class (mean -0.002, std 0.025) are all consistent with zero across seeds. Size is the only other rung with a seed-robust positive contribution (mean $+0.022$, std 0.005), an order of magnitude smaller than morphology's. We adopt the five-seed mean, $\Delta R^2_{\mathrm{morph}} = 0.09 \pm 0.03$, as the primary estimate throughout this paper. The morphology/pre-morphology ratio is $2.8 \pm 1.2\times$ (range 1.7–4.3×) across the same five seeds, the dominant single contribution under every seed tested. The identical protocol applied to the Galaxy Zoo Hubble ladder (Section 4.10) gives a Featured/Disk ΔR^2 of 0.153 ± 0.007 (range $[0.141, 0.160]$) – positive and tightly clustered under every seed, markedly more stable than the equivalent Zurich statistic.

4.1.2. *Redshift Sensitivity*

At the representative seed, inserting redshift after stellar mass, z alone contributes $\Delta R^2 = +0.108$, reducing $\Delta R^2_{\mathrm{morph}}$ to $+0.038$ (retention fraction 0.30 at this seed). Across five seeds this becomes 0.49 ± 0.24 (range $[0.28, 0.91]$; Table 2).

Applying the identical protocol to Galaxy Zoo Hubble (Section 4.10) gives 0.51 ± 0.03. A two-sample comparison of the means (standard error ≈ 0.107 for Zurich, ≈ 0.013 for GZ Hubble) gives $z \approx 0.19$ for a difference of 0.02: the two retention fractions are statistically indistinguishable, and both catalogues retain approximately half of their morphology signal after redshift control.

4.2. *Does Morphology Survive Removing AGN and Radio-Excess Hosts?*

The $\Delta R^2 \approx 0.09$ (five-seed mean) morphology gain is measured on the full sample, which includes AGN and radio-excess hosts. To test whether this gain reflects a property of star formation generally or is instead driven by the subset of sources where radio emission is AGN-contaminated, we rerun the ladder on two restricted subsamples across the same five seeds: sources with none of the three AGN flags set ($N = 3857$), and sources with the radio-excess flag unset ($N = 3840$).

Removing multiwavelength-flagged AGN leaves the morphology gain intact and seed-robust: mean $\Delta R^2 = +0.092$, std 0.029, range $[+0.043, +0.114]$, positive at every seed. Removing radio-excess hosts, however, removes the gain entirely and reverses its sign consistently: mean $\Delta R^2 = -0.008$, std 0.010, range $[-0.023, -0.001]$ – negative at all five seeds, though the magnitude is small relative to the seed-to-seed noise floor. The pre-morphology cumulative R^2 in this subsample is much higher to begin with (0.82 vs. 0.49 on the full sample at the reference seed), because radio-excess hosts are precisely the population that departs from the radio-SFR locus that luminosity, mass, size, and surface density already describe well; once they are excluded, morphology has no systematic outliers left to explain. Morphology's dominant contribution to the information ladder is identifying which sources are radio-excess/AGN hosts, not correcting calorimetric efficiency uniformly across the star-forming population.

4.3. *Mass Dependence and a Direct Surface-Density Test*

The calorimetric mechanism proposed for the residual morphology gain predicts that morphology should matter more for systems closer to the calorimetric limit – higher stellar mass and surface density – and should be more directly visible in SFR surface density than in SFR luminosity. We test both predictions.

4.3.1. *Mass Stratification*

Splitting the fixed ladder sample into stellar-mass terciles (median $\log(M_\star/M_\odot) = 10.43, 10.87, 11.22$; $N \approx 1610$–1618 per bin) and rerunning the ladder within each bin gives, at the reference seed, $\Delta R^2_{\rm morph} = +0.049, +0.136, +0.146$ respectively. Across five seeds the low-mass bin's effect is not robust: mean +0.010, std 0.031, range $[-0.029, +0.049]$, indistinguishable from zero and negative under some seeds. The mid- and high-mass bins are robust: mid mass is positive under every seed (mean +0.072, range $[+0.018, +0.136]$), and high mass is both large and stable (mean +0.186, std 0.044, range $[+0.146, +0.253]$). The morphology gain is undetectable above the noise floor at low stellar mass, and becomes a large, robust effect at high stellar mass – a threshold-like pattern consistent with a calorimetric-limit mechanism that engages once systems are close enough to the limit.

4.3.2. *Surface-Density Ratio and Permutation Importance*

The ratio $\log(\Sigma_{L,L_{\rm TIR,SF}}/\Sigma_\star)$ – SFR surface density normalised by stellar surface density – correlates with morphology far more strongly than SFR surface density alone: Spearman $\rho = -0.287$ (Gini), -0.345 (concentration), $+0.348$ (asymmetry), all with $p \lesssim 10^{-90}$; the raw SFR surface density alone gives $|\rho| \approx 0.06$–0.11 for Gini/concentration and is not significant for asymmetry at this precision. Compact, high-concentration, low-asymmetry hosts have systematically lower SFR output per unit existing stellar surface density. A grouped permutation-importance analysis on the full-feature MLP corroborates this attribution via a method that does not depend on rung order: the two radio luminosities dominate (importance 0.462, 0.423), the grouped mass/size/density unit ranks third (0.258), and concentration, asymmetry, and Gini follow individually (0.225, 0.194, 0.121) – summed, the three morphology features' importances (0.540) exceed the grouped mass/size/density unit's. Together with Section 4.2, these results indicate two distinct channels: an AGN/radio-excess identification channel, dominant in the raw ladder gain and requiring radio-excess hosts to be present, and a smaller but genuine calorimetric channel visible in the mass stratification and the surface-density ratio, operating within the star-forming population independent of AGN content.

4.4. *Independent AGN Classification Check*

As an independent cross-check on the multiwavelength AGN classification used throughout, we sky-match AllWISE photometry to the VLA-COSMOS sample (2.″0 tolerance; 3622/7729 sources matched) and apply the Stern et al.

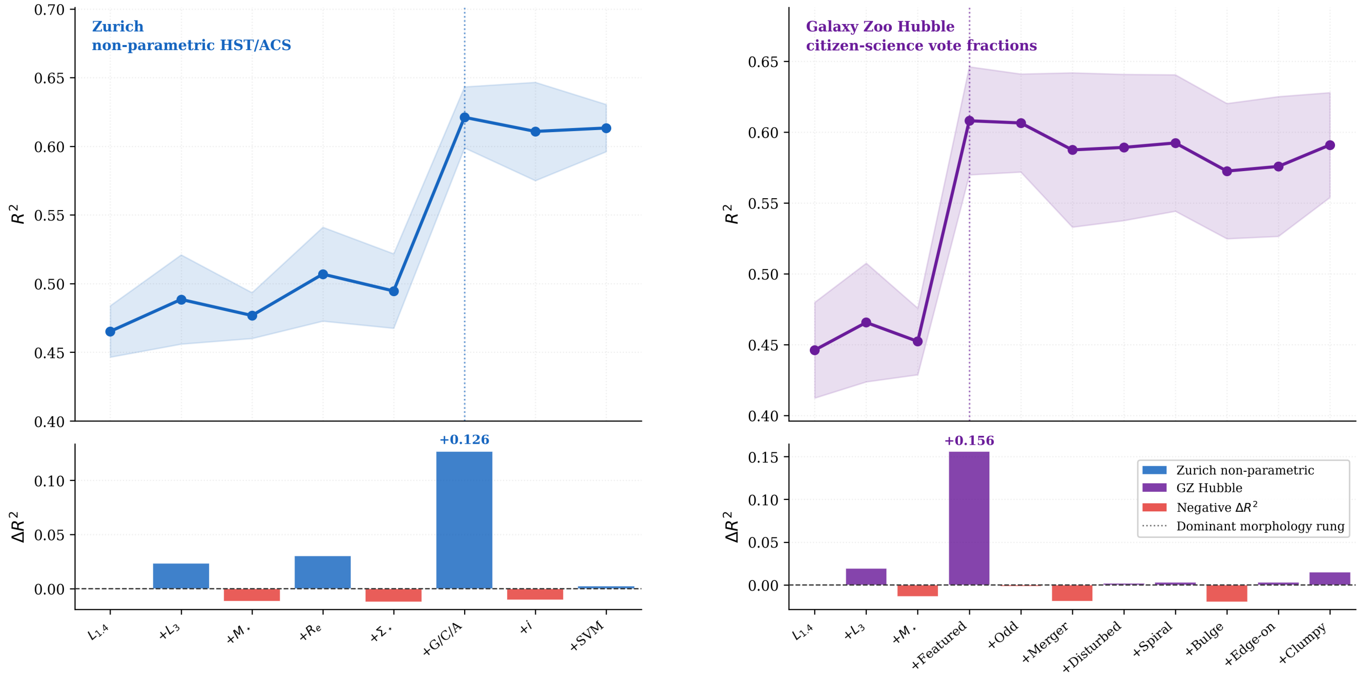

Figure 1. Information ladders for the two morphology systems (5-fold CV MLP; seed = 42; target: $\log L_{\rm TIR,SF}$). Left column: Zurich non-parametric HST/ACS morphology ($N = 4843$). Right column: Galaxy Zoo Hubble citizen-science vote fractions ($N = 3696$). Top panels show cumulative R^2 with $\pm1\sigma$ shading; bottom panels show incremental ΔR^2 bars. The featured/disk rung (right column, 4th bar) yields $\Delta R^2 = +0.156$ from a single volunteer vote fraction at this seed, comparable to the Zurich morphology step ($\Delta R^2 = +0.127$ at this seed, left column, 6th bar); the five-seed means are 0.153 ± 0.007 and 0.092 ± 0.028 respectively (Table 2).

(2012) $W1 - W2 > 0.8$ colour cut. The WISE-based flag agrees with the combined multiwavelength AGN flag for 75.8% of matched sources, supporting the classification used throughout without raising concerns about systematic optical/IR AGN misclassification in the flat-spectrum subsample (Section 4.6).

4.5. *Matched-Sample Morphological Tests*

Unique nearest-neighbour matching produces 1003 radio-excess and 1003 non-radio-excess control pairs (Zurich), and 697 pairs (Galaxy Zoo Hubble). Figure 4 presents the Cohen's d results across both systems.

Zurich full sample. Radio-excess hosts are more compact than controls: Gini $d = +0.28$ (95% CI $[+0.20, +0.38]$; $p = 4.5 \times 10^{-10}$); concentration $d = +0.27$ ($[+0.18, +0.36]$; $p = 1.5 \times 10^{-9}$). Asymmetry shows a significant negative offset: $d = -0.24$ ($[-0.33, -0.15]$; $p = 1.4 \times 10^{-10}$). Adding $\log L_{1.4\,\rm GHz}$ to the matched-control variables slightly strengthens these offsets relative to a mass-redshift-SFR-only match, confirming the signal is not a by-product of radio luminosity differences.

Galaxy Zoo Hubble full sample. Radio-excess hosts show a significantly lower featured/disk fraction ($d = -0.283$; $p = 2.0 \times 10^{-9}$) and the largest single-feature matched-sample effect in either system: a reduced spiral fraction ($d = -0.369$; 95% CI $[-0.478, -0.269]$; $p = 2.1 \times 10^{-8}$; Figure 7). The merger and disturbed fractions are also significant (Merger $d = +0.155$, $p = 0.036$; Disturbed $d = -0.151$, $p = 0.0026$), while the clumpy fraction is not ($d = +0.042$; $p = 0.159$). We discuss the physical interpretation of these six offsets together in Section 5.2.

Multiple-comparisons correction. Across the nine matched-sample tests reported for Zurich (three morphology metrics across the full, steep-spectrum, and flat-spectrum sub-samples), all nine remain significant after Holm–Bonferroni correction at family-wise $\alpha = 0.05$ (largest surviving raw $p = 0.0153$).

AUC by individual feature. The single-feature AUC values give a complementary view of the same population: spiral fraction (AUC $= 0.660 \pm 0.015$), featured/disk (AUC $= 0.620 \pm 0.020$), disturbed (AUC $= 0.575 \pm 0.023$), and merger (AUC $= 0.486 \pm 0.021$, slightly below chance). We discuss the interpretation of the merger AUC alongside its matched-sample offset in Section 5.2.

Table 3. Information Ladder and AUC: Zurich and Galaxy Zoo Hubble (Representative Seed, seed= 42)

Rung / Feature	System	N	ΔR^2 (MLP)	R^2 (MLP)	AUC
Zurich non-parametric (5-fold CV; seed = 42)					
Baseline: $L_{1.4\,\mathrm{GHz}}$	Zurich	4843	···	0.465	···
$+L_{3\,\mathrm{GHz}}$	Zurich	4843	+0.023	0.489	···
$+\log(M_\star/M_\odot)$	Zurich	4843	-0.012^a	0.477	···
$+R_e$	Zurich	4843	+0.030	0.507	···
$+\log\Sigma_\star$	Zurich	4843	-0.012^a	0.495	···
+Morphology (G, C, A)	Zurich	4843	**+0.127**	**0.621**	···
$+i_{\mathrm{proxy}}$	Zurich	4843	-0.010^a	0.611	···
+`CLASS_SVM`	Zurich	4843	+0.003	0.613	···
Redshift-sensitivity check (z after $\log(M_\star/M_\odot)$; MLP)					
$+z$ (rung 3b)	Zurich	4843	+0.108	0.603	···
+Morphology (with z controlled)	Zurich	4843	+0.038	0.652	···
Galaxy Zoo Hubble (5-fold CV; seed = 42)					
Baseline: $L_{1.4\,\mathrm{GHz}}$	GZ Hub.	3696	···	0.446	···
$+L_{3\,\mathrm{GHz}}$	GZ Hub.	3696	+0.020	0.466	···
$+\log(M_\star/M_\odot)$	GZ Hub.	3696	-0.013^a	0.452	···
+GZ: Featured/Disk	GZ Hub.	3696	**+0.156**	**0.608**	···
All subsequent GZ features	GZ Hub.	3696	$\leq \pm 0.020^a$	0.608	···
AUC (5-fold stratified logistic regression)					
Asymmetry	Zurich	4843	···	···	0.658 ± 0.016
All Zurich morphology (5 params.)	Zurich	4843	···	···	0.693 ± 0.009
Spiral fraction	GZ Hub.	3696	···	···	0.660 ± 0.015
All GZ Hubble features	GZ Hub.	3696	···	···	0.697 ± 0.009
Nested logistic model (Zurich; $N = 4843$)					
Base: $\log(M_\star/M_\odot) + z + \log L_{\mathrm{TIR,SF}}$	Zurich	4843	···	···	0.865 ± 0.015
Full: base + Gini, Conc., Asym.	Zurich	4843	···	···	0.865 ± 0.015

NOTE—[a] Negative ΔR^2 values reflect stochastic early-stopping convergence in the Adam optimiser, not genuine information loss. Values in this table are computed at a single representative seed (42), shown for direct comparison with Figure 1; the five-seed estimates in Table 2 are used throughout the rest of this paper. The nested-model BIC increases by +25.0 upon adding morphology despite identical AUC to three decimal places; a paired bootstrap over out-of-fold predictions (2000 resamples) gives $\Delta\mathrm{AUC} = -0.0002$, 95% CI $[-0.0007, +0.0002]$ (Section 4.8), which includes zero. This is expected under the two-channel picture (Section 4.2): the full-sample classifier is not sensitive to the mass/redshift/SFR/luminosity-controlled offset that the matched-sample test in Section 4.5 is specifically designed to detect.

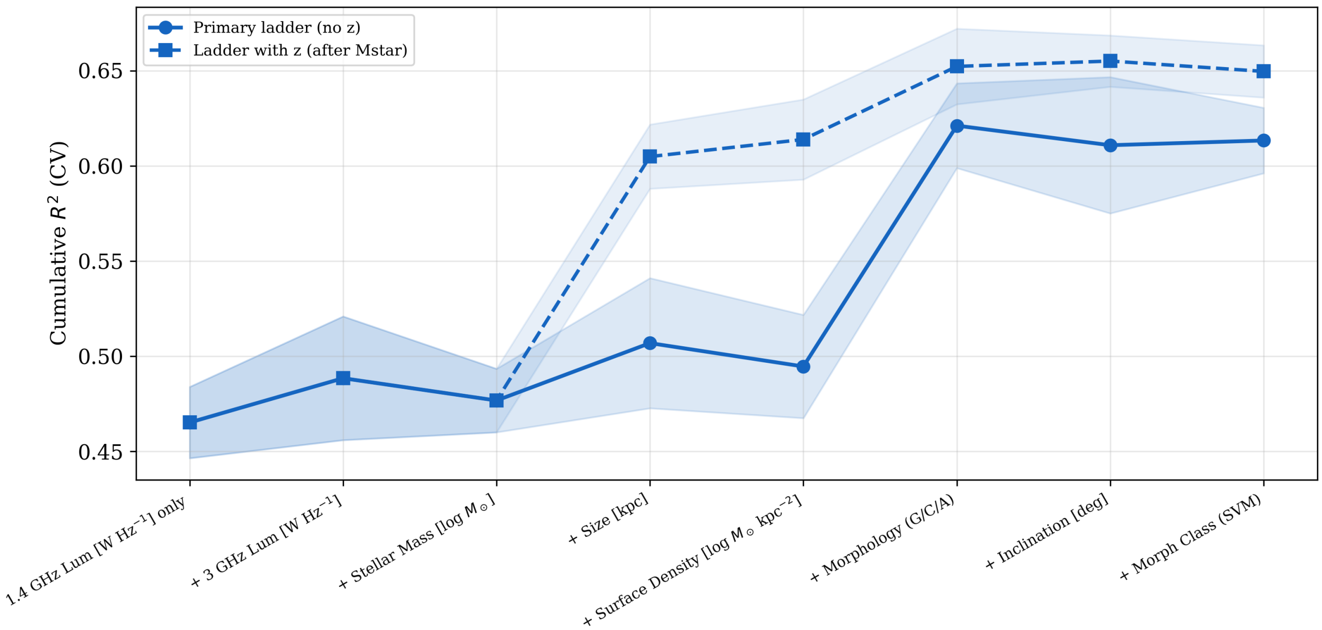

Figure 2. Redshift-sensitivity check on the Zurich information ladder (MLP; $N = 4843$; seed$= 42$). Solid circles trace the primary ladder without explicit redshift; open squares the ladder with redshift inserted after stellar mass. Shaded bands show $\pm 1\sigma$ from 5-fold cross-validation. Inserting z captures $\Delta R^2 = +0.108$ before the morphology rung, reducing $\Delta R^2_{\rm morph}$ from 0.127 to 0.038 at this seed; the five-seed retention fraction is 0.49 ± 0.24 (Section 4.1.2).

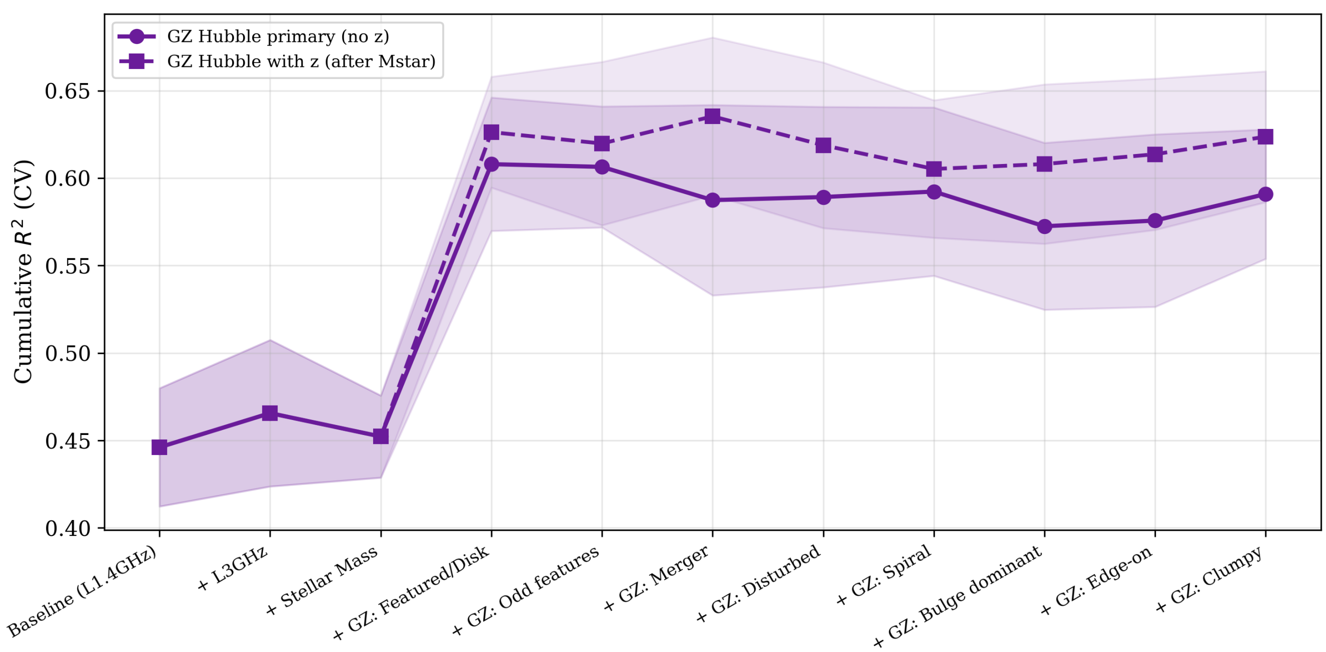

Figure 3. Redshift-sensitivity check on the Galaxy Zoo Hubble information ladder (MLP; $N = 3696$; seed$= 42$). Solid circles: primary ladder without explicit redshift. Dashed squares: ladder with redshift inserted after stellar mass. Inserting z captures $\Delta R^2 = +0.096$ before the featured/disk rung, reducing its ΔR^2 from $+0.156$ to $+0.078$ at this seed; the five-seed retention fraction is 0.51 ± 0.03 (Section 4.1.2).

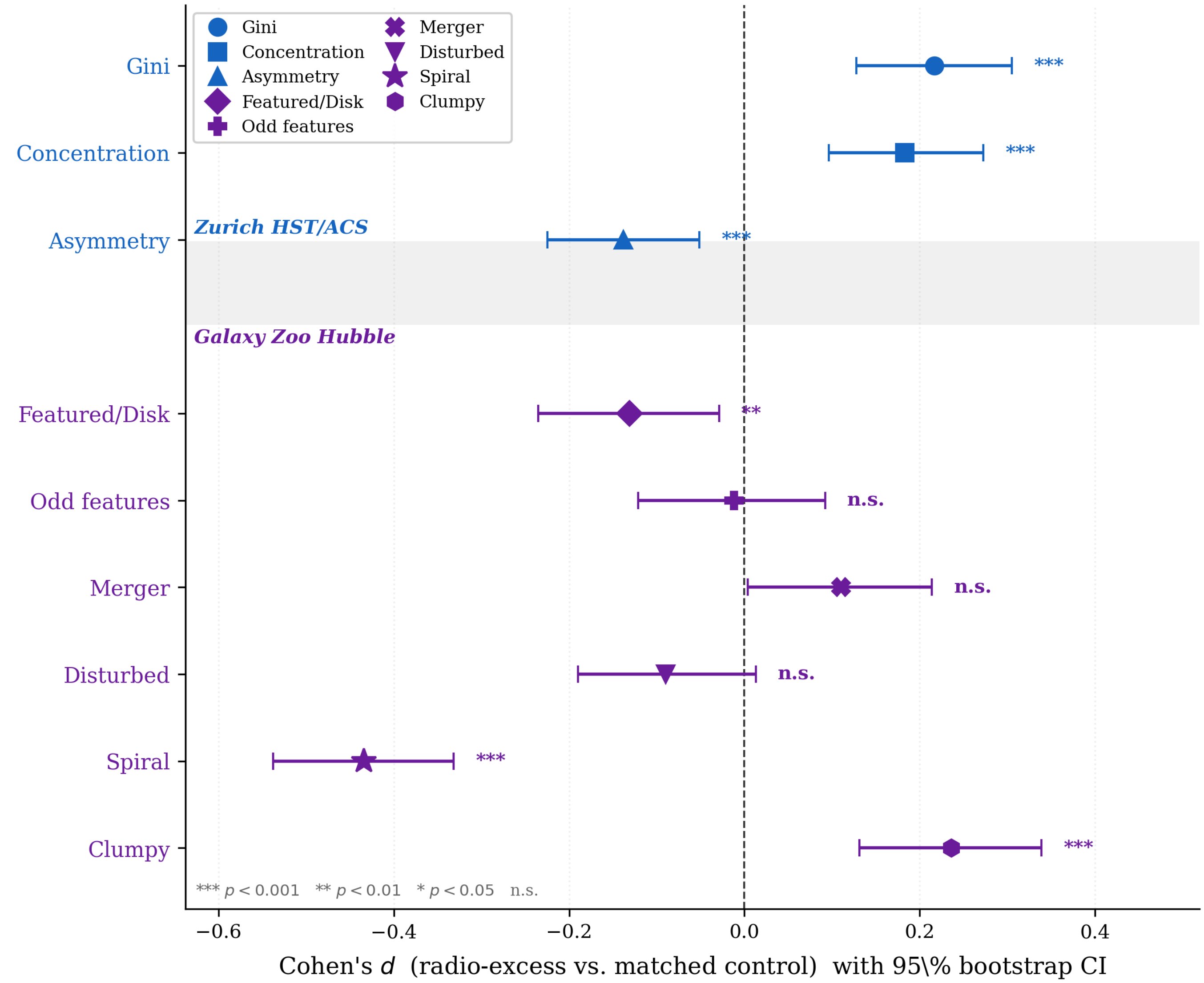

Figure 4. Matched-sample Cohen's d (radio-excess minus matched controls) from two independent morphology systems. Error bars are 95% bootstrap confidence intervals. Top block (blue): Zurich non-parametric ($N_{\rm rx} = 1003$ unique pairs; 5000 resamples); rows from top to bottom are Gini, Concentration, Asymmetry. Bottom block (purple): Galaxy Zoo Hubble ($N_{\rm rx} = 697$ unique pairs; 2000 resamples); rows from top to bottom are Featured/Disk, Odd features, Merger, Disturbed, Spiral, Clumpy (values in Table 4). The vertical dashed line marks $d = 0$. All pairs matched 1:1 in $(\log M_\star, z, \log L_{\rm TIR,SF}, \log L_{1.4\,\rm GHz})$; no replacement.

4.6. *The Sign Flip: Spectral Stratification Reveals Two Populations*

Splitting the Zurich matched sample by spectral index reveals a physically decisive differentiation (Table 4). Among steep-spectrum sources ($\alpha_{\rm radio} < -0.5$; $N_{\rm rx} = 864$), Gini ($d = +0.26$) and concentration ($d = +0.24$) retain their positive offsets; asymmetry shows a negative offset ($d = -0.19$; $p = 3.3 \times 10^{-7}$), consistent with a preference for smooth hosts in AGN populations after mass-and-SFR control (Cisternas et al. 2011; Mechtley et al. 2016). Among flat-spectrum sources ($\alpha_{\rm radio} \geq -0.5$; $N_{\rm rx} = 139$), the signal changes qualitatively and dramatically. Gini ($d = +0.28$) and concentration ($d = +0.35$) retain their positive offsets, while asymmetry shows a suppression of extraordinary magnitude:

$$d_{\rm Asym} = -0.867, \quad 95\%\ \mathrm{CI}\ [-1.114, -0.633], \quad p = 8.0 \times 10^{-13}. \tag{1}$$

The median asymmetry of radio-excess flat-spectrum hosts ($\langle A \rangle_{\rm rx} = 0.059$) is barely half that of their individually matched controls ($\langle A \rangle_{\rm non} = 0.110$). We treat this as suggestive of a genuine population-level effect given the modest

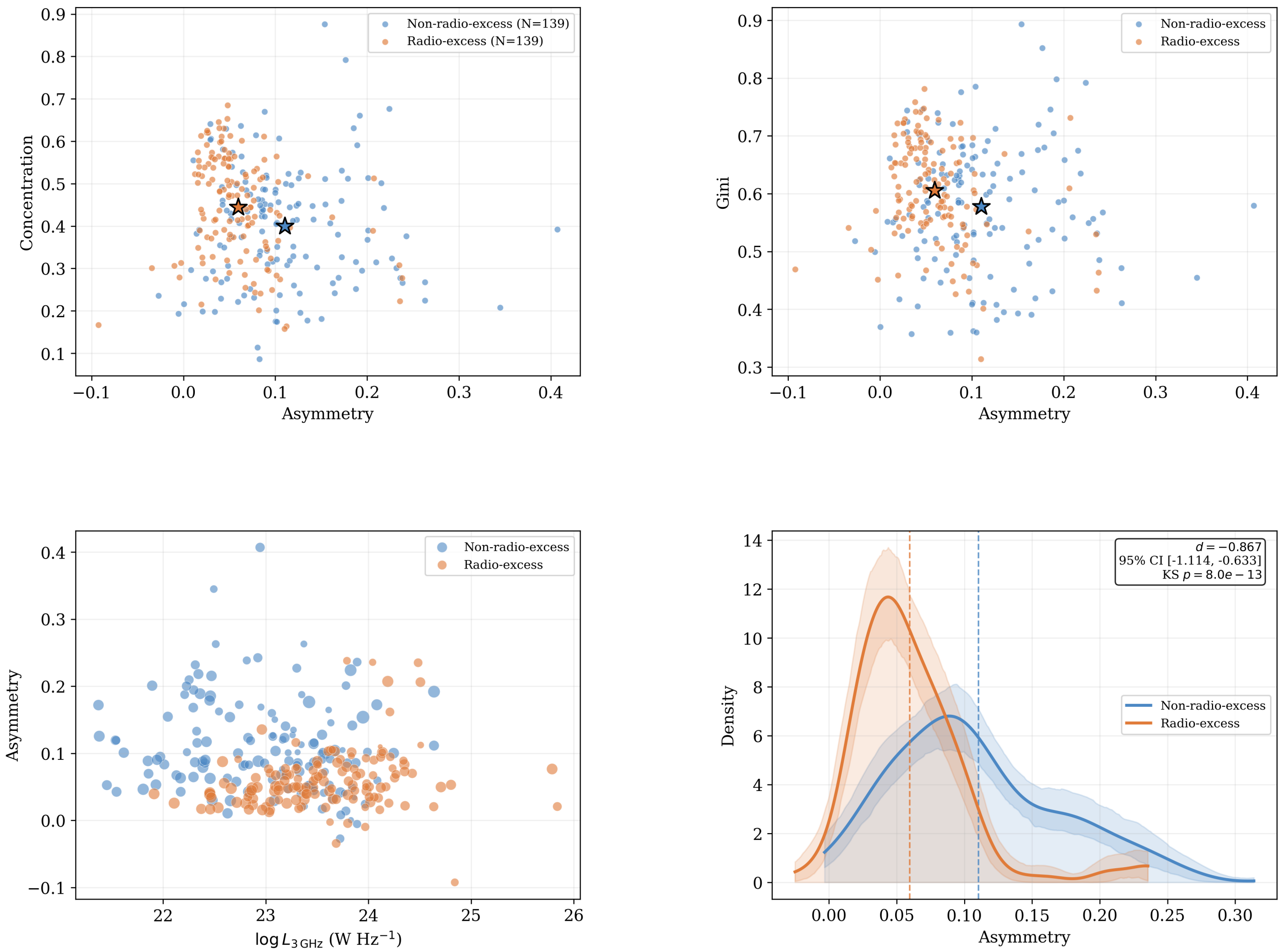

Figure 5. Individual source properties in the flat-spectrum sub-sample ($\alpha_{\mathrm{radio}} \geq -0.5$; $N = 278$). Orange points are radio-excess ($N_{\mathrm{rx}} = 139$), blue non-radio-excess ($N = 139$); stars mark group means. *Top left*: concentration vs. asymmetry; radio-excess sources cluster at low asymmetry across the full concentration range. *Top right*: Gini vs. asymmetry; asymmetry is the dominant separation axis. *Bottom left*: asymmetry vs. $\log L_{3\,\mathrm{GHz}}$ (point size proportional to Gini); no luminosity dependence of the asymmetry offset is apparent. *Bottom right*: 1D asymmetry KDEs with 95% bootstrap CI shading; the in-panel box gives Cohen's $d = -0.867$ ($p = 8 \times 10^{-13}$; $N_{\mathrm{rx}} = 139$ unique matched pairs; 5000 resamples).

sub-sample size ($N_{\mathrm{rx}} = 139$), pending independent confirmation at larger sample size, while noting that the result is directionally robust to spectral-index threshold cuts (Section 4.7). The radio-to-optical size ratio provides independent corroboration: across the full sample, radio-excess sources are more compact relative to their optical size ($d = +0.41$; $p = 1.3 \times 10^{-25}$).

4.7. *Robustness of the Flat-Spectrum Signal*

Both luminosities were derived under a fixed $\alpha = -0.7$, so the flat-spectrum sub-sample partly selects on scatter around that assumed value. Table 5 tests this systematically. The asymmetry suppression is unchanged at $\geq 1\sigma_\alpha$ ($d = -0.857$; $p = 1.7 \times 10^{-12}$; $N_{\mathrm{rx}} = 138$), remains highly significant at $\geq 1.5\sigma_\alpha$ ($d = -0.859$; $p = 4.6 \times 10^{-8}$; $N_{\mathrm{rx}} = 89$), and persists at $\geq 2\sigma_\alpha$ ($d = -0.711$; $p = 1.3 \times 10^{-4}$; $N_{\mathrm{rx}} = 56$), despite removing more than half the original sub-sample. A continuous test within the flat-spectrum radio-excess sub-sample finds no significant trend of asymmetry residual with spectral-index deviation (Spearman $\rho = +0.033$; $p = 0.70$): the suppression is a uniform population-level property, not a boundary artefact.

Table 4. Bootstrap Cohen's d: Zurich and Galaxy Zoo Hubble

Feature / Sub-sample	d	95% CI	KS p
Zurich non-parametric (1003 unique pairs; 5000 resamples)			
Full sample ($N_{\rm rx} = 1003$)			
Gini	+0.28	$[+0.20, +0.38]$	4.5×10^{-10}
Concentration	+0.27	$[+0.18, +0.36]$	1.5×10^{-9}
Asymmetry	−0.24	$[-0.33, -0.15]$	1.4×10^{-10}
Flat-spectrum ($\alpha_{\rm radio} \geq -0.5$; $N_{\rm rx} = 139$)			
Gini	+0.28	$[+0.05, +0.53]$	1.5×10^{-2}
Concentration	+0.35	$[+0.11, +0.59]$	1.0×10^{-2}
Asymmetry	**−0.87**	$[\mathbf{-1.11}, \mathbf{-0.63}]$	$\mathbf{8.0 \times 10^{-13}}$
Steep-spectrum ($\alpha_{\rm radio} < -0.5$; $N_{\rm rx} = 864$)			
Gini	+0.26	$[+0.17, +0.36]$	2.5×10^{-7}
Concentration	+0.24	$[+0.15, +0.34]$	9.6×10^{-7}
Asymmetry	−0.19	$[-0.28, -0.09]$	3.3×10^{-7}
Galaxy Zoo Hubble (697 unique pairs; 2000 resamples; full sample)			
Featured/Disk (debiased)	−0.283	$[-0.391, -0.176]$	2.0×10^{-9}
Odd features	−0.078	$[-0.179, +0.026]$	4.8×10^{-2}
Merger	+0.155	$[+0.052, +0.257]$	3.6×10^{-2}
Disturbed	−0.151	$[-0.257, -0.040]$	2.6×10^{-3}
Spiral	**−0.369**	$[\mathbf{-0.478}, \mathbf{-0.269}]$	$\mathbf{2.1 \times 10^{-8}}$
Clumpy	+0.042	$[-0.068, +0.148]$	1.6×10^{-1}

Note—All matches are unique one-to-one in $(\log(M_\star/M_\odot), z, \log L_{\rm TIR,SF}, \log L_{1.4\,\rm GHz})$ with no replacement. Zurich's nine tests (three metrics × three sub-samples) all survive Holm–Bonferroni correction at family-wise $\alpha = 0.05$ (largest surviving raw $p = 0.0153$).

Table 5. Robustness of Flat-Spectrum Asymmetry Suppression ($\sigma_\alpha = 0.20$)

Threshold	$N_{\rm rx}$	d	KS p	Status
Unrestricted	139	−0.867	8.0×10^{-13}	Reference
$\geq 0.5\sigma_\alpha$	139	−0.867	8.0×10^{-13}	Unchanged
$\geq 1.0\sigma_\alpha$	138	−0.857	1.7×10^{-12}	Persists
$\geq 1.5\sigma_\alpha$	89	−0.859	4.6×10^{-8}	Persists
$\geq 2.0\sigma_\alpha$	56	−0.711	1.3×10^{-4}	Persists

Note—Continuous test within the flat-spectrum radio-excess sub-sample: Spearman $\rho = +0.033$, $p = 0.70$.

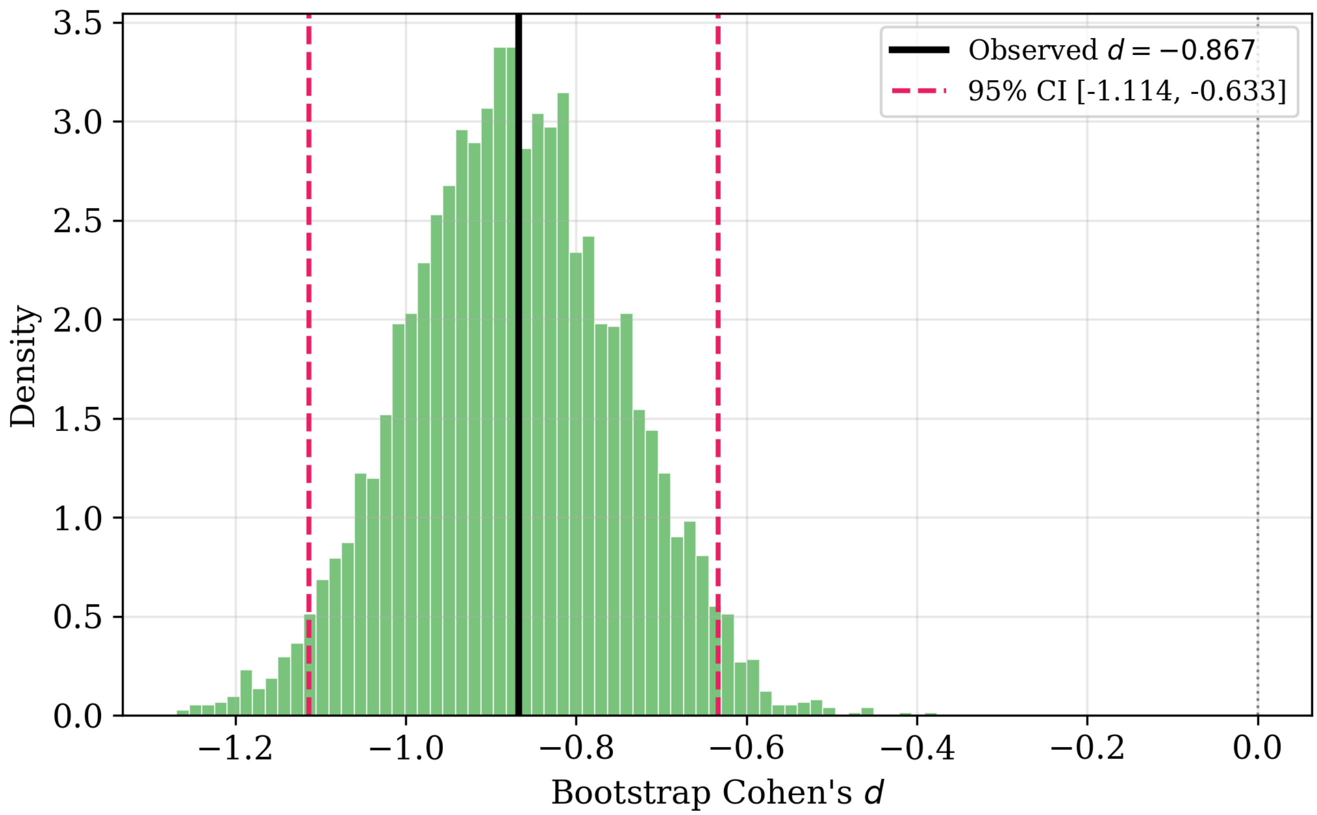

Figure 6. Bootstrap sampling distribution of Cohen's d for asymmetry in the flat-spectrum matched sub-sample ($\alpha_{\rm radio} \geq -0.5$; $N_{\rm rx} = 139$ unique matched pairs; 5000 resamples). The solid vertical line marks the observed $d = -0.867$; dashed lines mark the 95% CI $[-1.114, -0.633]$; the dotted line marks $d = 0$. All 5000 resampled d values lie below zero.

4.8. *Nested Logistic Model: A Null Result, Explained*

The base model ($\log(M_\star/M_\odot), z, \log L_{\rm TIR,SF}$) achieves AUC $= 0.865 \pm 0.015$; adding Gini, concentration, and asymmetry gives AUC $= 0.865 \pm 0.015$ – identical to three decimal places, despite a BIC increase of $+25.0$ for the full model. A paired bootstrap over out-of-fold predicted probabilities from the same cross-validation folds (2000 resamples) gives ΔAUC $= -0.0002$, 95% CI $[-0.0007, +0.0002]$, which includes zero. Adding morphology to a model already containing mass, redshift, and SFR produces no measurable improvement in full-sample radio-excess discrimination. This null result is expected under the two-channel picture developed in Section 5.3: the full-sample logistic classifier pools matched and unmatched pairs together and is not sensitive to the mass/redshift/SFR/luminosity-controlled offset that the matched-sample tests (Section 4.5) are specifically designed to detect. The matched-sample design, not the full-sample classifier, is the correct tool for this signal.

4.9. *Redshift Evolution and Completeness*

Both morphological cross-match completeness and morphology-only AUC decline monotonically across all four measured redshift bins for the Zurich system: AUC falls from 0.799 ± 0.046 at $z < 0.5$ to 0.575 ± 0.038 at $1.5 < z < 2.0$, while completeness drops from $\sim$85% to $\sim$38% over the same interval. In the broad-bin analysis, AUC declines from 0.748 ± 0.017 at $0 < z < 1$ to 0.622 ± 0.020 at $1 < z < 2$ (ΔAUC $= -0.127$; $p = 1.2 \times 10^{-5}$; Figure 8). For the Zurich system, the completeness trend alone is sufficient to explain the AUC decline without invoking structural evolution. The Galaxy Zoo Hubble system tells a different and more constraining story (Figure 9). GZH cross-match completeness is unity across all four redshift bins. Nevertheless, GZH AUC declines from 0.789 ± 0.029 at $z < 0.5$ to 0.570 ± 0.151 at $1.5 < z < 2.0$. Because completeness cannot explain this decline, it must reflect genuine measurement degradation in volunteer vote fractions at high redshift, despite the de-biasing correction, or genuine structural evo-

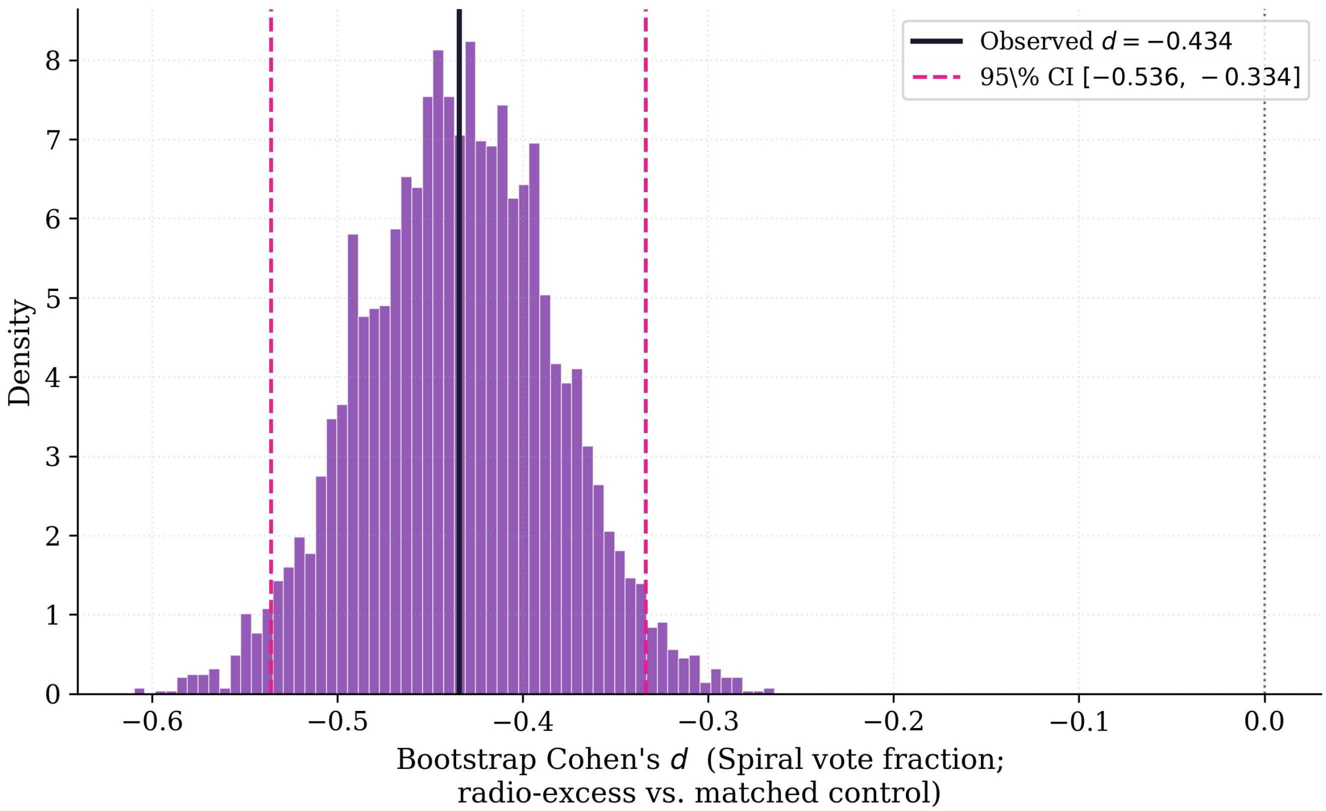

Figure 7. Bootstrap sampling distribution of Cohen's d for the Galaxy Zoo Hubble spiral vote fraction ($N_{\rm rx} = 697$ unique matched pairs; 2000 resamples; $d = -0.369$; Table 4). This is the largest single-feature matched-sample effect in the GZ Hubble system and the citizen-science counterpart of the Zurich asymmetry suppression.

lution of the radio-excess population. This is a stronger constraint than the Zurich result and merits follow-up with deeper space-based imaging to distinguish these two possibilities.

4.10. *Galaxy Zoo Hubble: Multi-Seed Extension*

Repeating the Galaxy Zoo Hubble information ladder across the same five seeds as the Zurich analysis gives a Featured/Disk ΔR^2 of 0.153 ± 0.007 (range $[0.141, 0.160]$), positive and tightly clustered under every seed – markedly more stable than the equivalent Zurich morphology-rung statistic. The retention-fraction result from this check is given in Section 4.1.2.

5. DISCUSSION

5.1. *Convergence Across Independent Systems*

AUC values of 0.693 and 0.697 from two instruments, algorithms, and classifier populations that share nothing except the galaxies observed are close enough to be notable, though this agreement alone is not so tight as to be implausible by chance; morphology-based AUCs in this range are not unusual in the AGN/radio-excess classification literature. The stronger evidence for a real physical signal is that both systems (i) show a mass-stratified pattern consistent with a genuine mechanism rather than a single coincidental number (Section 4.3), and (ii) behave consistently under the same multi-seed and matched-sample scrutiny: their retention fractions agree with each other (Section 4.1.2), which is itself a form of convergence.

5.2. *A Decision Tree for Morphological Contributions to Radio-SFR Calibration*

The three analyses (ladder, matched-sample tests, and single-feature AUC) ask different questions and are dominated by different morphological features. Bringing them together as a decision tree clarifies what each structural measurement contributes and in which observational context it should be applied.

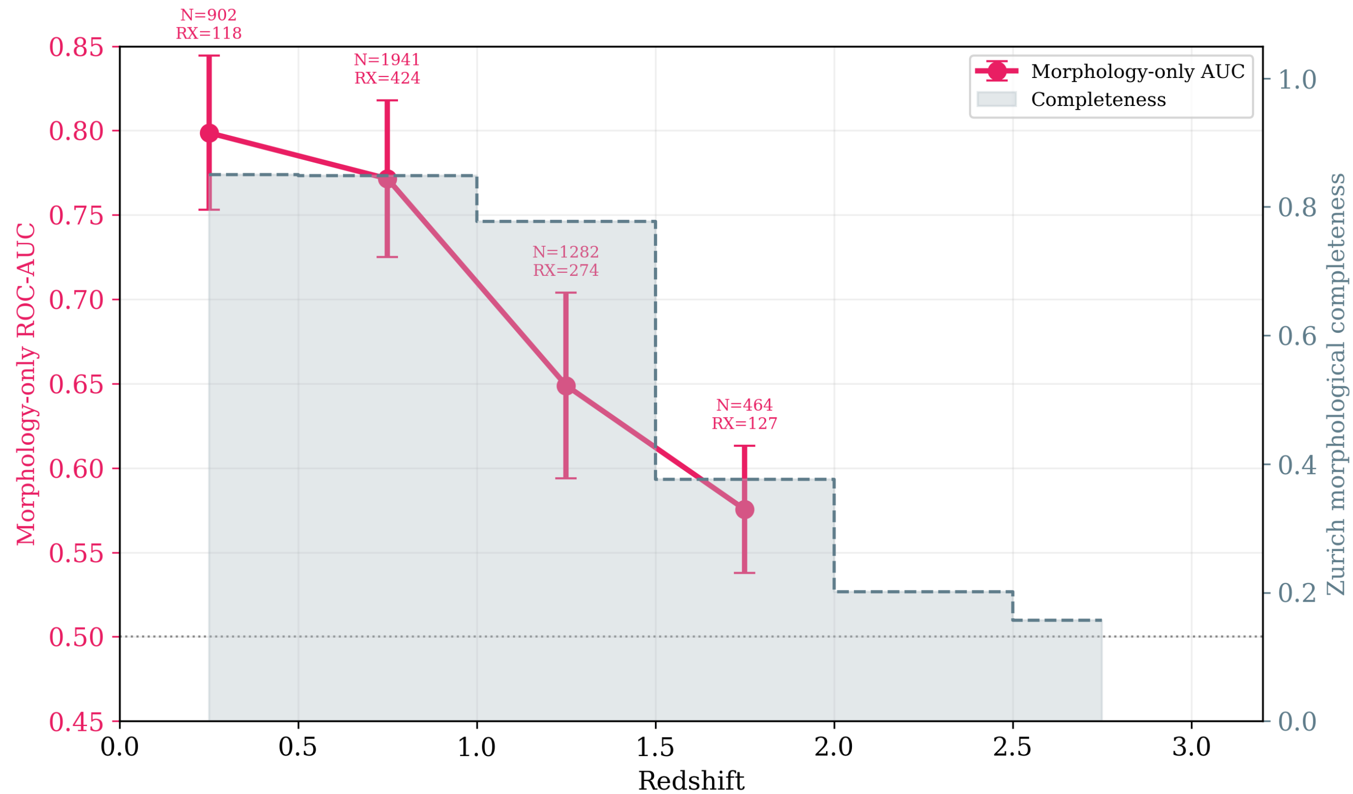

Figure 8. Zurich non-parametric morphology-only ROC-AUC (left axis, pink circles with error bars) and morphological cross-match completeness (right axis, gray shading) as a function of redshift in $\Delta z = 0.5$ bins. AUC declines from 0.799 ± 0.046 at $z < 0.5$ to 0.575 ± 0.038 at $1.5 < z < 2.0$; completeness declines from $\sim$85% to $\sim$38% over the same interval. The dotted line marks AUC = 0.5.

Branch 1: Improving the radio-SFR calibration for the full population. Use the featured/disk vote fraction (or its pixel-level equivalent, Gini, concentration, and asymmetry). The smooth/featured axis encodes calorimetric variation across the population, most clearly among massive, high-surface-density hosts (Section 4.3). The morphology rung delivers $\Delta R^2 = 0.09$–0.15 (five-seed means, Table 2) depending on the system; no subsequent feature adds meaningful variance.

Branch 2: Identifying which galaxies host radio-excess AGN. Use the spiral vote fraction (single-feature AUC = 0.660 ± 0.015) or, in the Zurich system, concentration and Gini offsets ($d \approx +0.27$) and the negative asymmetry offset ($d = -0.24$). Radio-excess hosts are significantly less likely to be spiral-structured than mass-and-SFR-matched controls ($d = -0.369$; $p = 2.1 \times 10^{-8}$; Table 4), the largest single-feature matched-sample effect in either system.

Branch 3: Identifying secular versus merger-triggered AGN within radio-excess hosts. The merger and disturbed vote fractions are individually significant in the matched-sample test (Merger $d = +0.155$, $p = 0.036$; Disturbed $d = -0.151$, $p = 0.0026$); the merger AUC nonetheless remains at chance level (0.486 ± 0.021) in the full unmatched sample – a different test, sensitive to rank-ordering across the full mass-redshift distribution rather than a mean shift in matched pairs. We read this combination as mixed evidence for merger-triggering rather than a clean rule-out or confirmation. The clumpy fraction is not significant ($d = +0.042$, $p = 0.159$) and does not currently support a structural claim in either direction. Among flat-spectrum sources, the Zurich asymmetry suppression ($d_{\mathrm{Asym}} = -0.867$; $p = 8 \times 10^{-13}$) is extreme and points toward secular fuelling – bar-driven inflows, minor unresolved interactions, or cold-mode accretion – rather than gas-rich major mergers, though we present this as suggestive given the small sub-sample (Section 4.6).

5.3. *Two Physical Channels*

Sections 4.2 and 4.3 show the raw morphology gain is not a single effect. Two distinct channels are responsible.

Channel 1: AGN/radio-excess identification (dominant). Most of the raw morphology gain disappears, and reverses sign, when radio-excess hosts are removed (Section 4.2). Gini and concentration are proxies for com-

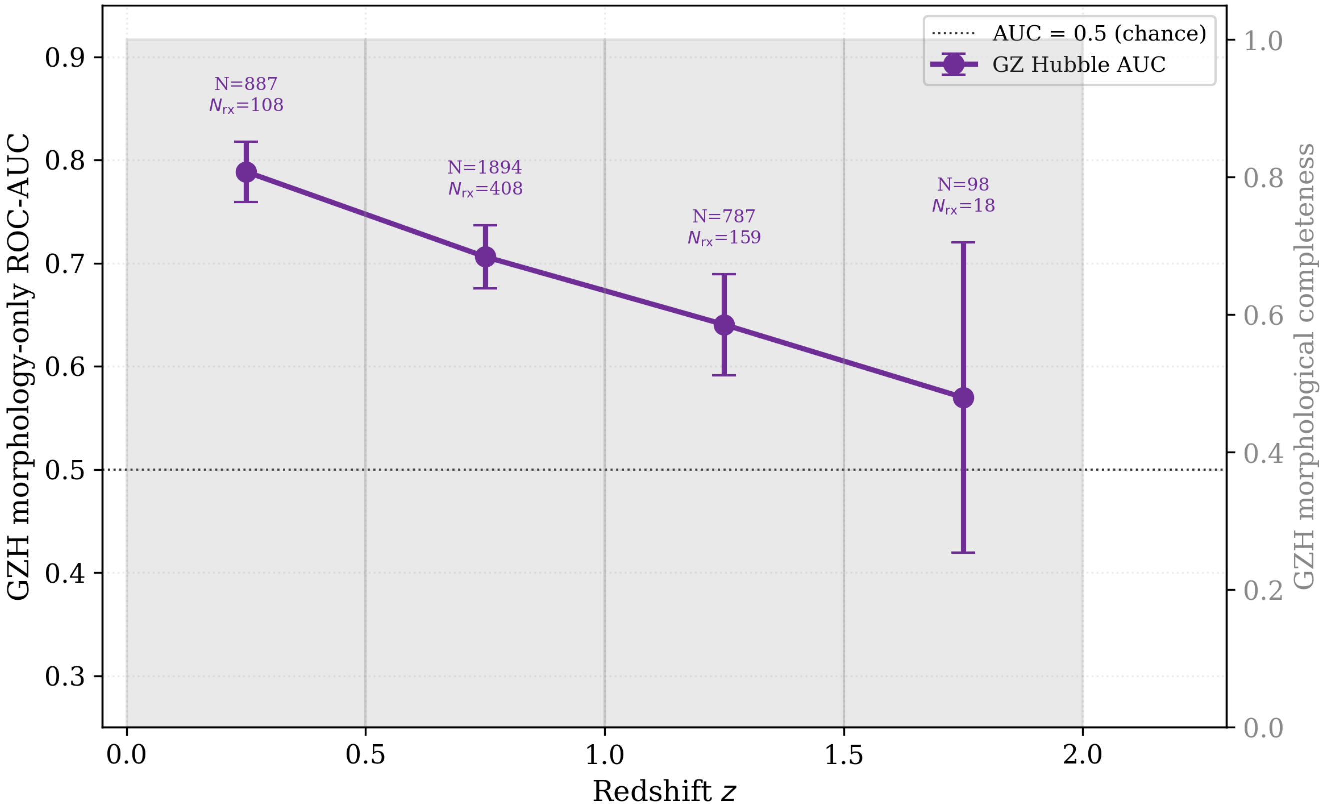

Figure 9. Galaxy Zoo Hubble morphology-only ROC-AUC (left axis, purple circles with error bars) and GZH cross-match completeness (right axis, gray shading) as a function of redshift in $\Delta z = 0.5$ bins. GZH cross-match completeness is unity across all four redshift bins; the AUC nevertheless declines from 0.789 ± 0.029 at $z < 0.5$ to 0.570 ± 0.151 at $1.5 < z < 2.0$. The dotted line marks AUC = 0.5.

pactness, and asymmetry for structural disturbance; radio-excess hosts are more compact and smoother than mass/redshift/SFR/luminosity-matched controls (Section 4.5). Morphology's main job in the information ladder is flagging this subpopulation, not correcting a universal calorimetric relation.

Channel 2: calorimetric regulation among star-forming hosts (smaller but genuine). Once AGN and radio-excess hosts are set aside, a secondary effect remains, visible in two independent tests. First, the morphology gain is strongly mass-dependent, undetectable at low mass and large and robust at high mass (Section 4.3), consistent with morphology mattering more as systems approach the calorimetric limit (Völk 1989; Lacki et al. 2010). Second, the ratio of SFR surface density to stellar surface density correlates with morphology 3–6× more strongly than SFR surface density alone – a more direct observational signature of a calorimetric effect than the luminosity-based ladder rung captures, corroborated by the grouped permutation-importance analysis.

The asymmetry suppression among flat-spectrum radio-excess hosts (Section 4.6) sits within Channel 1: it is a property of the AGN identification signal, concentrated in a small ($N_{\rm rx} = 139$) subsample, consistent with a preference for undisturbed hosts among this population rather than confirmed evidence for any specific AGN fuelling mode (Section 5.4).

5.4. *Physical Interpretation of the Flat-Spectrum Population*

The extreme asymmetry suppression ($d = -0.87$; $p = 8 \times 10^{-13}$) in flat-spectrum radio-excess hosts is the most striking single number in this work. The median asymmetry of these hosts is barely half that of their individually matched controls, and the suppression persists through $\geq 2\sigma_\alpha$ threshold cuts. Because it is derived from a matched-sample comparison rather than a dynamical or spectroscopic measurement, it is evidence of association between flat-spectrum radio excess and undisturbed host morphology, not direct evidence of a specific fuelling channel. One interpretation consistent with the pattern is secular fuelling (bar-driven inflows, minor interactions too subtle to imprint morphological asymmetry, or cold-mode accretion) rather than gas-rich major mergers, consistent with independent studies of confirmed GPS/CSS hosts (Holt 2009; O'Dea & Saikia 2021). We note this as the most physically plausible reading given the data in hand, not as a confirmed mechanism; direct confirmation would need resolved radio morphology (VLBI, e-MERLIN) and integral-field spectroscopy.

5.5. *Survey Implications for the SKA Era*

Three practical conclusions follow from the two-channel picture. First, optical morphology from Euclid, Rubin LSST, or Roman will deliver a substantial improvement to radio-excess/AGN screening ($\Delta R^2 \simeq 0.09$, five-seed mean, beyond radio luminosity and stellar mass), concentrated among intermediate- and high-mass hosts and among the radio-excess population specifically, rather than uniformly across all star-forming galaxies. Makechemu et al. (2026) establish that modelling flexibility alone cannot close this gap; morphological information is necessary. Second, within the star-forming population itself, structure still matters, but more modestly and mass-dependently; calibrators should not assume a flat morphological correction applies at all masses. Third, the Galaxy Zoo Hubble result establishes that a coarse smooth/featured classification captures most of the Channel 1 signal with a stability across training seeds that exceeds the pixel-level Zurich metrics, suggesting citizen-science-style classification may be preferable for large-scale automated screening even where precision imaging is available. Automated classifiers at citizen-science quality, already demonstrated at scale (Dieleman et al. 2015), are a natural route to this. Fourth, flat-spectrum radio-excess sources with smooth morphologies and high concentration form a physically distinct class worth treating separately in any morphology-based survey classification scheme, pending confirmation at larger sample size.

6. CONCLUSIONS

We have quantified the hierarchy of host-galaxy information governing radio emission as a star-formation tracer across three independent morphological measurement systems applied to VLA-COSMOS 3 GHz sources ($0 < z \lesssim 3$; median $z = 0.91$ for the primary $N = 4843$ sample), applying a multi-seed robustness protocol throughout. Our conclusions are:

1. Non-parametric morphology is a genuine, seed-robust contributor to the radio-SFR ladder: $\Delta R^2 = 0.09 \pm 0.03$ (Zurich, five-seed mean), exceeding the seed-to-seed noise floor of every other rung by 5σ.

2. This gain decomposes into two channels: a dominant AGN/radio-excess identification channel (the gain vanishes and reverses sign when radio-excess hosts are removed) and a smaller, genuine calorimetric channel that is undetectable at low stellar mass and becomes large and robust at high stellar mass, corroborated by a surface-density-ratio test and an order-independent permutation-importance analysis.

3. The Galaxy Zoo Hubble citizen-science catalogue independently replicates the signal (combined AUC 0.697 ± 0.009 vs. 0.693 ± 0.009) with a Featured/Disk ΔR^2 more stable across training seeds than the equivalent Zurich metric, and a redshift-sensitivity retention fraction (0.51 ± 0.03) statistically indistinguishable from Zurich's (0.49 ± 0.24).

4. Radio-excess hosts are compact and morphologically undisturbed at fixed mass, redshift, SFR, and radio luminosity in both systems: all nine matched-sample tests reported for Zurich survive correction for multiple comparisons, and the Galaxy Zoo Hubble matched sample shows a significant spiral deficit ($d = -0.369$), the largest single-feature effect in either system, alongside significant merger and disturbed offsets and a non-significant clumpy fraction.

5. Flat-spectrum radio-excess hosts show extreme asymmetry suppression ($d = -0.87$) consistent with secular AGN triggering, persisting through spectral-index threshold cuts; given this subsample's small size, we present this as suggestive pending independent confirmation.

6. A single smooth/featured classification from Euclid, Rubin LSST, or Roman will deliver a substantial improvement to radio-excess/AGN screening in the SKA era, and a smaller, mass-dependent improvement to radio-SFR calibration within the star-forming population.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

This work is based entirely on public catalogues. The VLA-COSMOS 3 GHz Large Project catalogue (Smolčić et al. 2017) and the associated multiwavelength counterpart and SED-derived properties (Delvecchio et al. 2017) are available through the COSMOS archive; the Zurich/Tasca HST/ACS morphology catalogue (Tasca et al. 2009) and the Galaxy Zoo Hubble vote-fraction catalogue (Willett et al. 2017) are available from their respective published sources. The analysis pipeline that reproduces the information ladders, matched-sample tests, and all figures in this paper, including the multi-seed robustness scripts described in Section 3.1.1, is available from the corresponding author on reasonable request and will be deposited in a public repository upon acceptance.

Software: Astropy (Astropy Collaboration et al. 2013, 2018, 2022), scikit-learn (Baron 2019), statsmodels

REFERENCES

Abraham, R. G., van den Bergh, S., & Nair, P. 2003, ApJ, 588, 218, doi: 10.1086/373919

Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167, doi: 10.3847/1538-4357/ac7c74

Astropy Collaboration, Price-Whelan, A. M., Sipőcz, B. M., et al. 2018, AJ, 156, 123, doi: 10.3847/1538-3881/aabc4f

Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, A&A, 558, A33, doi: 10.1051/0004-6361/201322068

Bamford, S. P., et al. 2009, MNRAS, 393, 1324, doi: 10.1111/j.1365-2966.2008.14252.x

Baron, D. 2019, Frontiers in Astronomy and Space Sciences, 6, 57, doi: 10.3389/fspas.2019.00057

Bell, E. F. 2003, ApJ, 586, 794, doi: 10.1086/367829

Cisternas, M., Jahnke, K., Inskip, K. J., et al. 2011, ApJ, 726, 57, doi: 10.1088/0004-637X/726/1/57

Condon, J. J. 1992, ARA&A, 30, 575, doi: 10.1146/annurev.aa.30.090192.003043

Conselice, C. J. 2003, ApJS, 147, 1, doi: 10.1086/375001

Del Moro, A., Alexander, D. M., Mullaney, J. R., et al. 2013, A&A, 549, A59, doi: 10.1051/0004-6361/201219880

Delhaize, J., Smolčić, V., Delvecchio, I., et al. 2017, A&A, 602, A4, doi: 10.1051/0004-6361/201629430

Delvecchio, I., Smolčić, V., Zamorani, G., et al. 2017, A&A, 602, A3, doi: 10.1051/0004-6361/201629367

—. 2021, A&A, 647, A123, doi: 10.1051/0004-6361/202040113

Dieleman, S., Willett, K. W., & Dambre, J. 2015, MNRAS, 450, 1441, doi: 10.1093/mnras/stv632

Ellison, S. L., Mendel, J. T., Patton, D. R., & Scudder, J. M. 2013, MNRAS, 435, 3627, doi: 10.1093/mnras/stt1562

Helou, G., Soifer, B. T., & Rowan-Robinson, M. 1985, ApJL, 298, L7, doi: 10.1086/184556

Holt, J. 2009, in Proceedings of Science: The Low-Frequency Radio Universe. https://arxiv.org/abs/0812.2812

Lacki, B. C., Thompson, T. A., & Quataert, E. 2010, ApJ, 717, 1, doi: 10.1088/0004-637X/717/1/1

Laigle, C., McCracken, H. J., Ilbert, O., et al. 2016, ApJS, 224, 24, doi: 10.3847/0067-0049/224/2/24

Lintott, C. J., et al. 2008, MNRAS, 389, 1179, doi: 10.1111/j.1365-2966.2008.13689.x

Lotz, J. M., Jonsson, P., Cox, T. J., & Primack, J. R. 2008, MNRAS, 391, 1137, doi: 10.1111/j.1365-2966.2008.14004.x

Lotz, J. M., Primack, J., & Madau, P. 2004, AJ, 128, 163, doi: 10.1086/421849

Madau, P., & Dickinson, M. 2014, ARA&A, 52, 415, doi: 10.1146/annurev-astro-081811-125615

Magnelli, B., Ivison, R. J., Lutz, D., et al. 2015, A&A, 573, A45, doi: 10.1051/0004-6361/201424937

Makechemu, J. S., Chibueze, J. O., & Simmons, B. D. 2026, ApJ

Mechtley, M., Jahnke, K., Windhorst, R. A., et al. 2016, ApJ, 830, 156, doi: 10.3847/0004-637X/830/2/156

Murphy, E. J., Condon, J. J., Schinnerer, E., et al. 2011, ApJ, 737, 67, doi: 10.1088/0004-637X/737/2/67

Novak, M., Smolčić, V., Delhaize, J., et al. 2017, A&A, 602, A5, doi: 10.1051/0004-6361/201629436

O'Dea, C. P., & Saikia, D. J. 2021, A&A Rev., 29, 3, doi: 10.1007/s00159-021-00131-w

Planck Collaboration, Aghanim, N., et al. 2020, A&A, 641, A6, doi: 10.1051/0004-6361/201833910

Schleicher, D. R. G., & Beck, R. 2013, A&A, 556, A142, doi: 10.1051/0004-6361/201321707

Schober, J., Schleicher, D. R. G., & Klessen, R. S. 2016, ApJ, 827, 109, doi: 10.3847/0004-637X/827/2/109

Scoville, N., Aussel, H., Benson, A., et al. 2007, ApJS, 172, 1, doi: 10.1086/516585

Smolčić, V., Novak, M., Bondi, M., et al. 2017, A&A, 602, A1, doi: 10.1051/0004-6361/201628704

Stern, D., et al. 2012, ApJ, 753, 30, doi: 10.1088/0004-637X/753/1/30

Tasca, L. A. M., Kneib, J.-P., Iovino, A., et al. 2009, A&A, 503, 379, doi: 10.1051/0004-6361/200811538

Thompson, T. A., Quataert, E., & Murray, N. 2006, ApJ, 645, 186, doi: 10.1086/504035

Völk, H. J. 1989, A&A, 218, 67

Willett, K. W., et al. 2013, MNRAS, 435, 2835, doi: 10.1093/mnras/stt1458

—. 2017, MNRAS, 464, 4176, doi: 10.1093/mnras/stw2568

Yun, M. S., Reddy, N. A., & Condon, J. J. 2001, ApJ, 554, 803, doi: 10.1086/321380